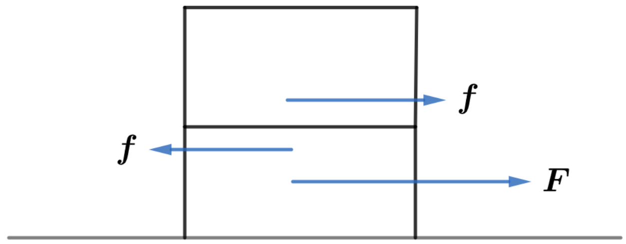
2020A F=ma Exam: Problem 14

Kevin S. Huang



If the two blocks stay together, then we have

$$F = 2ma$$
$$a = \frac{F}{2m}$$

Considering the top block, we have

$$f = ma = \frac{F}{2}$$

We know $f < \mu_s N = \mu_s mg$ so

$$\frac{F}{2} < \mu_s mg$$
$$F < 2\mu_s mg$$

is the condition for the two blocks to stay together. If F goes above this threshold, then $f = \mu_k N = \mu_k mg$ and then the acceleration of bottom block is given by

$$F - f = F - \mu_k mg = ma$$
$$a = \frac{F}{m} - \mu_k g$$

Thus,

$$a = \begin{cases} \frac{F}{2m} & F < 2\mu_s mg \\ \frac{F}{m} - \mu_k g & F > 2\mu_s mg \end{cases}$$

so the answer is $\boxed{\text{B}}$.